

zomatoproject

May 24, 2026

Zomato Data Analysis Project

Step 1 - Importing Libraries

pandas is used for data manipulation and analysis. numpy is used for numerical operations. matplotlib.pyplot and seaborn are used for data visualization.

```
[41]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Step-2 Create the data frame

```
[42]: dataframe = pd.read_csv("Zomato data .csv")
```

```
[43]: print(dataframe['rate'].head())
```

```
0    4.1/5
1    4.1/5
2    3.8/5
3    3.7/5
4    3.8/5
```

Name: rate, dtype: object

```
[44]: def handleRate(value):
    value = str(value).split('/')[0]

    if value == 'NEW' or value == '-':
        return None

    return float(value)
```

```
[45]: dataframe['rate'] = dataframe['rate'].apply(handleRate)
```

```
[46]: print(dataframe['rate'].head())
```

```
0    4.1
1    4.1
2    3.8
3    3.7
```

4 3.8

Name: rate, dtype: float64

```
[47]: dataframe
```

```
[47]:
```

	name	online_order	book_table	rate	votes	\
0	Jalsa	Yes	Yes	4.1	775	
1	Spice Elephant	Yes	No	4.1	787	
2	San Churro Cafe	Yes	No	3.8	918	
3	Addhuri Udupi Bhojana	No	No	3.7	88	
4	Grand Village	No	No	3.8	166	
..	...	...	...	...		
143	Melting Melodies	No	No	3.3	0	
144	New Indraprasta	No	No	3.3	0	
145	Anna Kuteera	Yes	No	4.0	771	
146	Darbar	No	No	3.0	98	
147	Vijayalakshmi	Yes	No	3.9	47	

	approx_cost(for two people)	listed_in(type)
0	800	Buffet
1	800	Buffet
2	800	Buffet
3	300	Buffet
4	600	Buffet
..	...	...
143	100	Dining
144	150	Dining
145	450	Dining
146	800	Dining
147	200	Dining

[148 rows x 7 columns]

Type of Resturant

```
[48]: dataframe.head()
```

```
[48]:
```

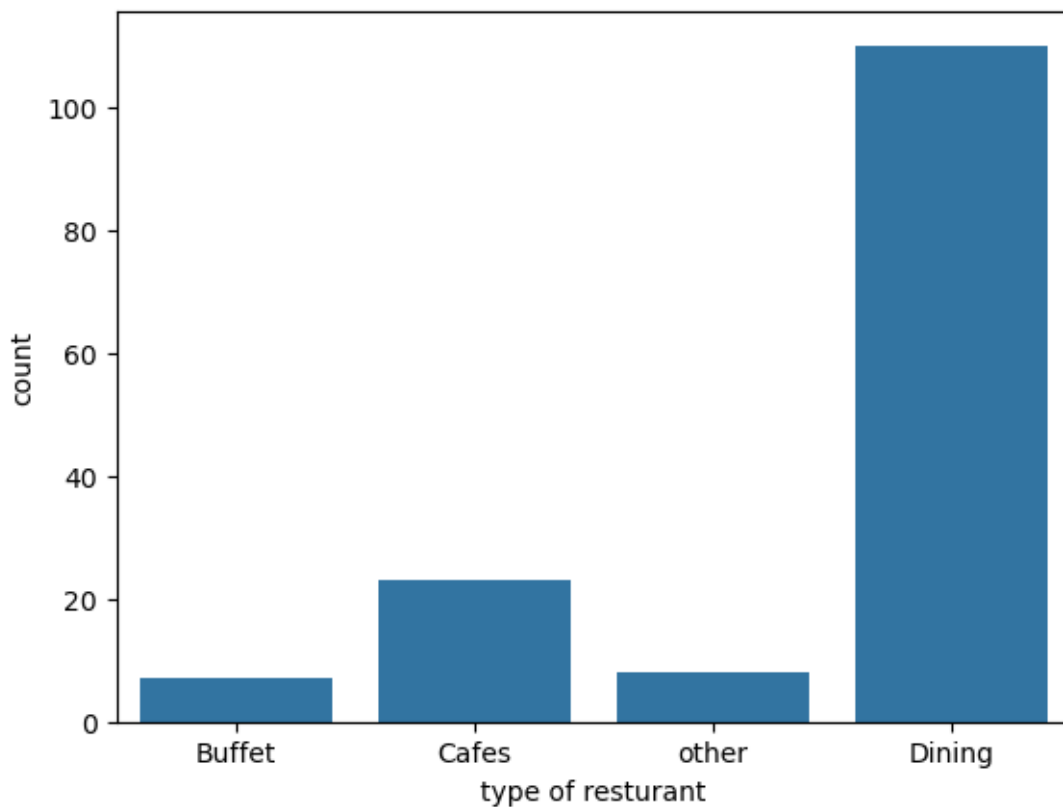
	name	online_order	book_table	rate	votes	\
0	Jalsa	Yes	Yes	4.1	775	
1	Spice Elephant	Yes	No	4.1	787	
2	San Churro Cafe	Yes	No	3.8	918	
3	Addhuri Udupi Bhojana	No	No	3.7	88	
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	approx_cost(for two people)	listed_in(type)
0	800	Buffet
1	800	Buffet
2	800	Buffet

3	300	Buffet
4	600	Buffet

```
[51]: sns.countplot(x=dataframe['listed_in(type)'])
plt.xlabel("type of resturant")
```

```
[51]: Text(0.5, 0, 'type of resturant')
```



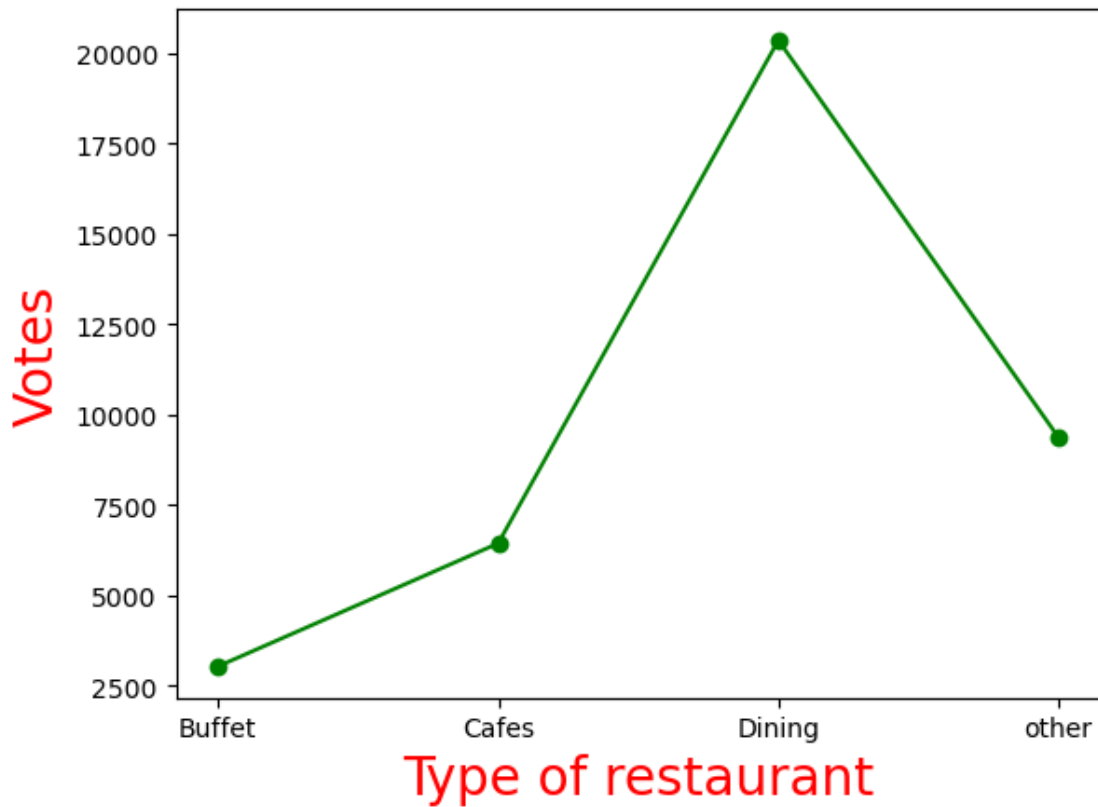
Conclusion: The majority of the restaurants fall into the dining category.

Dining restaurants are preferred by a larger number of individuals.

```
[52]: grouped_data = dataframe.groupby('listed_in(type)')['votes'].sum()
result = pd.DataFrame({'votes': grouped_data})

plt.plot(result, c="green", marker="o")
plt.xlabel("Type of restaurant", c="red", size=20)
plt.ylabel("Votes", c="red", size=20)
```

```
[52]: Text(0, 0.5, 'Votes')
```



Conclusion - Dining resturant has received maximum votes.

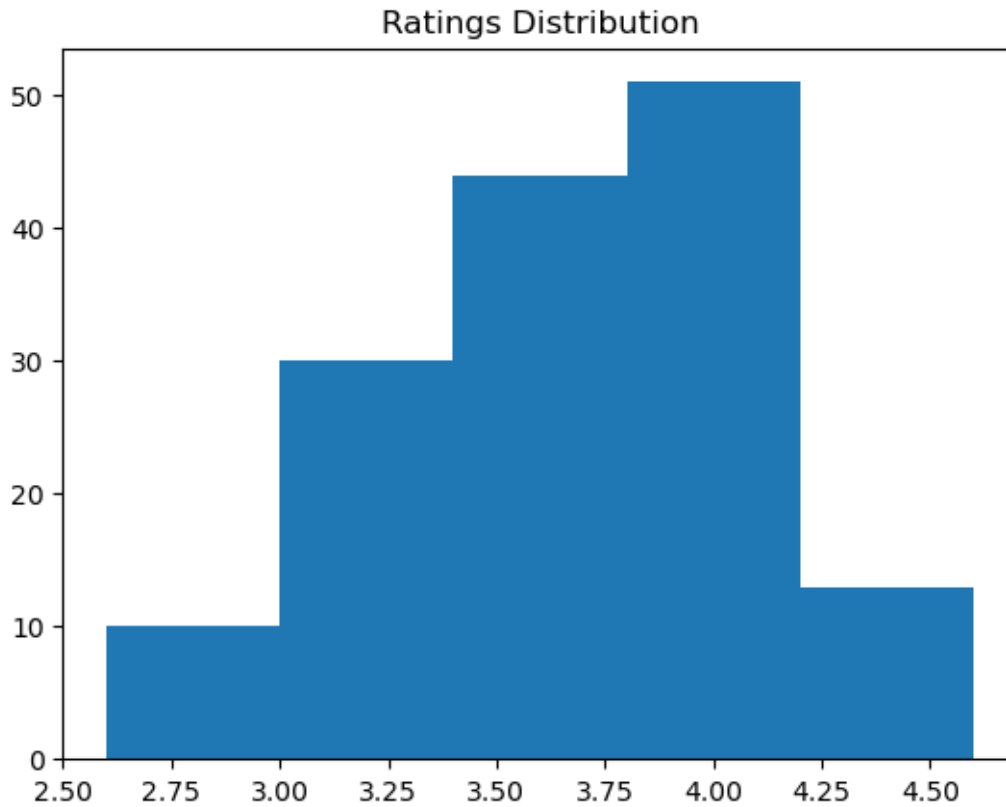
```
[59]: dataframe.head()
```

```
[59]:
```

	name	online_order	book_table	rate	votes	\
0	Jalsa	Yes	Yes	4.1	775	
1	Spice Elephant	Yes	No	4.1	787	
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	approx_cost(for two people)	listed_in(type)
0	800	Buffet
1	800	Buffet
2	800	Buffet
3	300	Buffet
4	600	Buffet

```
[53]: plt.hist(dataframe['rate'],bins=5)
plt.title("Ratings Distribution")
plt.show()
```

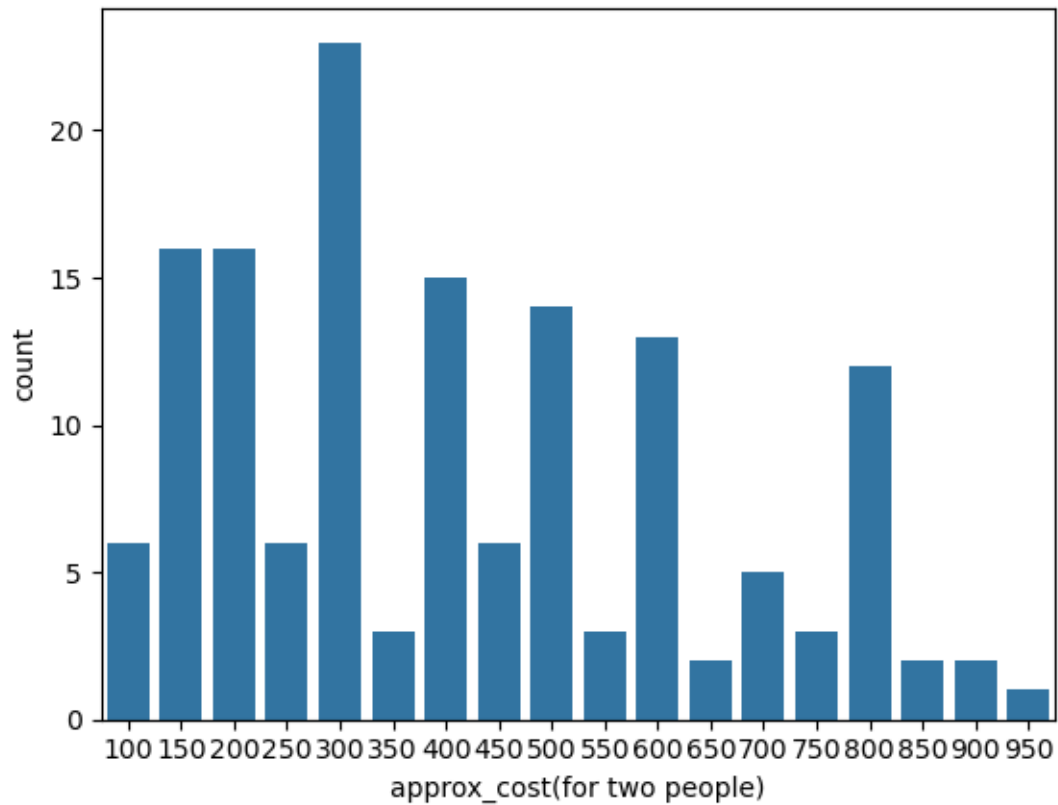


Conclusion: The majority of restaurants received ratings ranging from 3.5 to 4.

The majority of couples prefer restaurants with an approximate cost of 300 rupees.

```
[54]: couple_data=dataframe['approx_cost(for two people)']  
sns.countplot(x=couple_data)
```

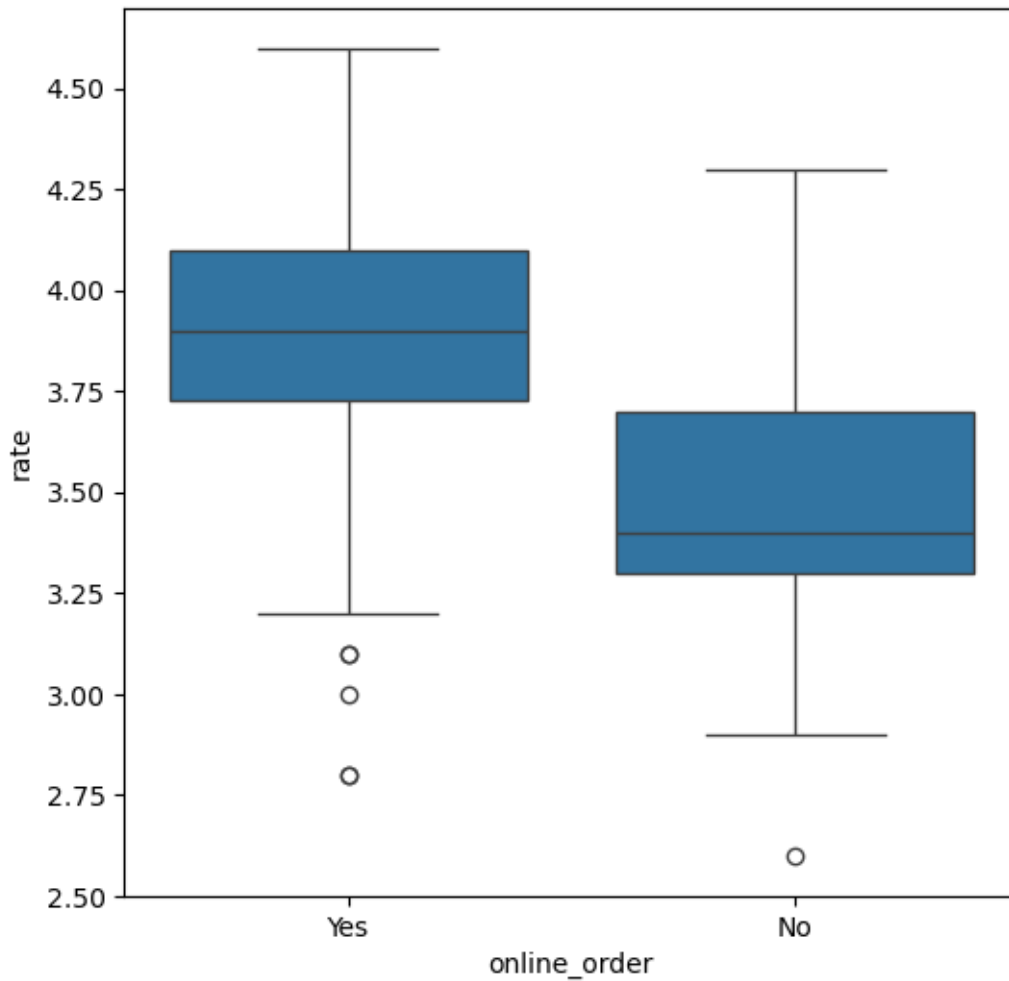
```
[54]: <Axes: xlabel='approx_cost(for two people)', ylabel='count'>
```



whether online orders receive higher ratings than offline orders.

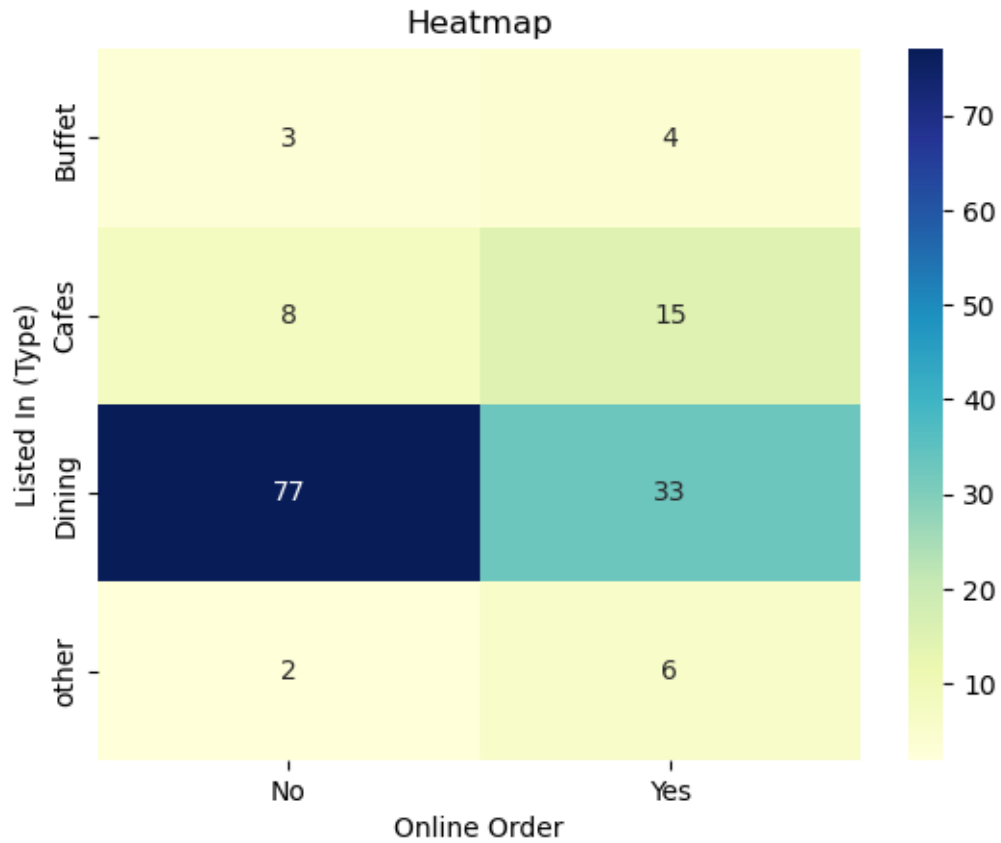
```
[55]: plt.figure(figsize = (6,6))
      sns.boxplot(x= 'online_order', y = 'rate', data = dataframe)
```

```
[55]: <Axes: xlabel='online_order', ylabel='rate'>
```



CONCLUSION: offline orders received lower ratings in comparison to online orders, which obtained excellent ratings.

```
[58]: pivot_table = dataframe.pivot_table(  
    index='listed_in(type)',  
    columns='online_order',  
    values='votes',  
    aggfunc='count'  
)  
  
sns.heatmap(pivot_table, annot=True, cmap="YlGnBu", fmt='.0f')  
  
plt.title("Heatmap")  
plt.xlabel("Online Order")  
plt.ylabel("Listed In (Type)")  
plt.show()
```



CONCLUSION : Dining restaurants primarily accept offline orders , whereas cafes primarily receive online orders. This suggests that clients prefer to place orders in person at restaurants, but prefer online ordering at cafes.